INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Vantage WA13, WA -- station KEPH, America/Los_Angeles
Committed pair	OSM 1193914601 -> 283051471 OSM way 1193914601 / OSM way 283051471
Facade gap	75.2 m between the two halls
Weather record	42,965 real hourly records from KEPH
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.2860 C at 220 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-02-02 (knife_edge)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 11 of 23 hours with 1 mode change. The reactive incumbent took 23 hours -- 12 more -- but declared 0 unsafe hours against the agent's 0. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	11 of 23 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	23 of 23 hours free cooling, 1 mode change(s), 0 unsafe hour(s) 11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	4.566	18.0	4.713	5.165
01	mechanical	yes	switch budget	4.561	18.0	5.144	4.361
02	mechanical	yes	switch budget	4.598	18.0	5.246	3.564
03	mechanical	yes	switch budget	4.609	18.0	4.136	3.349
04	mechanical	yes	switch budget	4.572	18.0	3.560	3.098
05	mechanical	yes	switch budget	5.125	18.0	4.710	3.390
07	mechanical	yes	switch budget	7.386	18.0	4.136	2.898
08	mechanical	yes	switch budget	10.661	18.0	5.270	3.365
09	mechanical	yes	switch budget	13.461	18.0	6.151	4.032
10	mechanical	yes	switch budget	15.153	18.0	8.050	5.012
11	mechanical	yes	switch budget	17.002	18.0	9.160	5.809
12	mechanical	no	dry-bulb	18.140	18.0	10.251	5.384

13	FREE	yes	-	16.271	18.0	8.391	4.957
14	FREE	yes	-	15.166	18.0	8.615	4.126
15	FREE	yes	-	14.562	18.0	7.913	4.032
16	FREE	yes	-	13.458	18.0	6.356	3.096
17	FREE	yes	-	12.953	18.0	4.596	3.394
18	FREE	yes	-	12.053	18.0	2.506	3.163
19	FREE	yes	-	8.758	18.0	1.815	3.428
20	FREE	yes	-	7.545	18.0	2.430	3.946
21	FREE	yes	-	6.802	18.0	1.396	4.645
22	FREE	yes	-	5.763	18.0	-0.610	5.433
23	FREE	yes	-	4.086	18.0	-0.575	5.241

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.566 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.561 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.598 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.610 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.572 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.125 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.387 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.661 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.461 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.153 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.002 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.140 C against a 18.0 C limit, so it fails by 0.140 C. It would take a limit 0.140 C higher, or a bound 0.140 C tighter, to change this hour.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.271 C, which is 1.729 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.957 C, and the margin is measured, not chosen: 4.806 C of group-conditional forecast error for this hour of day, plus 0.1514 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.167 C, which is 2.833 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.126 C, and the margin is measured, not chosen: 3.960 C of group-conditional forecast error for this hour of day, plus 0.1659 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.562 C, which is 3.438 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.032 C, and the margin is measured, not chosen: 3.910 C of group-conditional forecast error for this hour of day, plus 0.1219 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.458 C, which is 4.542 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.096 C, and the margin is measured, not chosen: 2.978 C of group-conditional forecast error for this hour of day, plus 0.1176 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.953 C, which is 5.047 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.394 C, and the margin is measured, not chosen: 3.226 C of group-conditional forecast error for this hour of day, plus 0.1676 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.053 C, which is 5.947 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.163 C, and the margin is measured, not chosen: 2.983 C of group-conditional forecast error for this hour of day, plus 0.1802 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.758 C, which is 9.242 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.428 C, and the margin is measured, not chosen: 3.243 C of group-conditional forecast error for this hour of day, plus 0.1845 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.545 C, which is 10.455 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.946 C, and the margin is measured, not chosen: 3.804 C of group-conditional forecast error for this hour of day, plus 0.1418 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.802 C, which is 11.198 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.645 C, and the margin is measured, not chosen: 4.513 C of group-conditional forecast error for this hour of day, plus 0.1318 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.763 C, which is 12.237 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.433 C, and the margin is measured, not chosen: 5.260 C of group-conditional forecast error for this hour of day, plus 0.1725 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.086 C, which is 13.914 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.241 C, and the margin is measured, not chosen: 5.103 C of group-conditional forecast error for this hour of day, plus 0.1373 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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